

Cisplatin Timing in Oral and Head-and-Neck Cancer: Organ-Specific Toxicity Mechanisms, Clinical Delivery Constraints, and the Rationale for Chronotherapy

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Abstract

Cisplatin-based concurrent chemoradiotherapy is the standard of care for locally advanced oral squamous cell carcinoma (OSCC) and head and neck squamous cell carcinoma (HNSCC), but nephrotoxicity, ototoxicity, oral mucositis, nausea and vomiting, and peripheral neuropathy limit dose delivery in a substantial proportion of patients. These toxicities are organ-specific: each reflects a distinct biology of transporter-mediated platinum uptake, DNA adduct accumulation, repair capacity, and tissue-specific apoptotic response. An emerging body of molecular evidence indicates that several determinants of cisplatin organ toxicity — including OCT2-mediated renal tubular uptake, XPA-dependent nucleotide excision repair activity, glutathione redox buffering, and mucosal epithelial proliferative rate — oscillate with circadian periodicity under the control of the molecular clock. Mechanistic studies, a small OSCC pilot study, and a phase II randomised trial in nasopharyngeal carcinoma support the hypothesis that the circadian time of cisplatin administration modulates toxicity outcomes; however, no chronotherapy study in head-and-neck cancer has been powered for survival, and the sex×timing interaction documented in colorectal cancer trials demands sex-stratified design. This review synthesises the organ-specific toxicity mechanisms, circadian timing gates, and clinical evidence relevant to OSCC/HNSCC. We propose a chronotype-aware phase II trial framework in which the primary endpoint is Grade ≥ 3 mucositis or Grade ≥ 3 nephrotoxicity within six weeks of first cisplatin, with mandatory sex stratification and prospective circadian function index (CFI) monitoring, as the most defensible near-term translational step.

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1. Introduction

Oral squamous cell carcinoma (OSCC) and the broader category of head and neck squamous cell carcinomas (HNSCC) collectively affect more than 900,000 patients each year and carry a five-year survival rate of approximately 50% for locally advanced disease despite
5 advances in surgical technique and intensity-modulated radiotherapy [1]. Cisplatin has occupied the central chemotherapeutic position in these cancers for more than four decades, forming the backbone of concurrent chemoradiotherapy for definitive and postoperative treatment of stage III–IV disease and of platinum-based induction regimens for selected patients. Its continued predominance reflects both demonstrated survival benefit and the
10 absence of a well-tolerated oral or parenteral alternative of equivalent efficacy.

Cisplatin’s effectiveness is constrained by a narrow therapeutic window. At the doses required for radiosensitisation — typically 100 mg m^{-2} every three weeks or 40 mg m^{-2} weekly — cisplatin consistently causes nephrotoxicity, high-frequency sensorineural hearing loss, oral mucositis, nausea and vomiting, peripheral neuropathy, and myelosuppression.
15 Clinically significant acute kidney injury occurs in approximately 20–30% of patients receiving high-dose cisplatin under contemporary hydration protocols, and cochlear platinum retention persists for months to years after the final treatment cycle [2]. Separately, up to 40% of patients enrolled in landmark postoperative chemoradiotherapy trials were unable to complete all three planned high-dose cisplatin cycles [3, 4]. These are not
20 merely discomforting side-effects: dose reductions, treatment interruptions, and premature cisplatin discontinuation each independently compromise locoregional disease control and survival.

The distribution of cisplatin toxicity across organs is not random. Each affected tissue accumulates platinum through a specific transporter biology, sustains DNA damage at a
25 characteristic rate governed by local repair capacity, and responds through tissue-specific apoptotic cascades. The kidney is vulnerable because OCT2-mediated influx concentrates platinum in proximal tubular cells; the cochlea because platinum accumulates in the stria vascularis and organ of Corti in cells that do not regenerate; the oral mucosa because high epithelial turnover maximises S-phase DNA damage; and dorsal root ganglia

neurons because they accumulate cisplatin without the protection of the blood–brain barrier. Understanding these organ-specific mechanisms is a prerequisite for any rational timing-based intervention.

An emerging body of evidence indicates that several determinants of cisplatin toxicity — notably OCT2 expression in renal tubular cells, the rate of nucleotide excision repair mediated by XPA, and the glutathione redox buffering capacity — oscillate with circadian periodicity under the control of the molecular clock. If these oscillations are of sufficient amplitude and phase-coherence in vivo, the time of cisplatin administration within the 24-hour cycle could systematically shift the balance between platinum accumulation, DNA repair capacity, and apoptotic threshold in normal tissues — offering a route to toxicity reduction without dose modification and without the biological cost of a dose reduction.

This review synthesises the evidence bearing on this hypothesis in the context of OSCC/HNSCC. We describe the clinical role of cisplatin in treatment pathways (Section 3), the organ-specific mechanisms of its principal toxicities (Section 4), the circadian timing gates most relevant to those mechanisms (Section 5), and the current clinical and preclinical evidence for cisplatin chronotherapy (Section 6). We then propose a chronotype-aware trial framework in which toxicity reduction — not survival improvement — is the primary research hypothesis (Section 7). The core argument is that cisplatin toxicity in OSCC/HNSCC is not merely dose-dependent, but organ-specific, pathway-specific, and potentially time-dependent, and that chronotype-aware cisplatin scheduling represents a testable, mechanistically plausible strategy for reducing the toxicity burden that limits treatment completion in this population.

2. Cisplatin chemistry and DNA damage: a timing-sensitive injury cascade

Cisplatin (*cis*-diamminedichloroplatinum(II)) is a square-planar platinum(II) coordination compound whose cytotoxic activity depends fundamentally on its *cis* geometry: the *trans* isomer fails to cross-link DNA and has no clinical antitumour activity [1]. At the high chloride concentrations found in plasma (approximately 100 mmol L^{−1}), cisplatin is relatively stable. Once inside cells, where intracellular chloride activity is substantially lower, sequential hydrolytic displacement of the two chloride ligands generates reactive mono- and di-aquated platinum species [5]. These electrophilic intermediates have high affinity for nucleophilic sites on DNA, and the intracellular platinum burden that results

from this activation is governed by the balance among cellular uptake, sequestration, detoxification, and efflux.

2.1. Cellular uptake and efflux transporters

The copper transporter CTR1 (SLC31A1) is the principal influx route for cisplatin in most cell types [6]. Of particular clinical relevance is the organic cation transporter 2, OCT2 (SLC22A2), highly expressed at the basolateral membrane of renal proximal tubular cells and in cochlear stria vascularis cells [5]: OCT2 mediates high-capacity platinum accumulation specifically in these tissues, which explains why cisplatin — unlike carboplatin or oxaliplatin — preferentially damages the kidney and cochlea [5, 2]. Intracellular platinum is exported by the copper-transporting P-type ATPases ATP7A and ATP7B; in OSCC, ATP7B overexpression is the dominant acquired resistance mechanism [6]. All three transport axes are documented or plausible targets of circadian regulation (Section 5).

2.2. DNA adduct formation

Once activated, aquated cisplatin reacts preferentially with the N7 positions of purine bases, forming covalent platinum-DNA adducts. The dominant lesion is the 1,2-intrastrand d(GpG) crosslink, in which two adjacent guanine residues on the same DNA strand are bridged by a single platinum atom [7]. Additional adducts include 1,2-d(ApG) intrastrand crosslinks and, less frequently, 1,3-intrastrand and interstrand adducts [7]. All platinum-DNA adducts introduce local helix distortion that is efficiently recognised by the DNA damage surveillance machinery [8]. The 1,2-d(GpG) crosslink and the 1,3-d(GpXpG) intrastrand diadduct are the principal cisplatin lesions processed by nucleotide excision repair, and they are the adducts responsible for both the cytotoxic efficacy of cisplatin and the DNA-damage burden that drives normal-tissue injury when repair is overwhelmed.

2.3. Nucleotide excision repair, checkpoint signalling, and apoptosis

Nucleotide excision repair (NER) is the sole mammalian pathway for removing bulky platinum-DNA adducts [8]. XPA serves as the damage-verification scaffold, recruiting the incision complex (XPC, TFIIH, RPA, ERCC1-XPF/XPG) [8, 9]; XPA abundance is the rate-limiting determinant of NER capacity and, critically, oscillates with robust circadian periodicity (Section 5). When adduct burden exceeds repair capacity, ATR–CHK1/ATM–CHK2 checkpoint cascades stabilise p53, which transactivates the pro-apoptotic effector

PUMA- α . PUMA- α liberates BAX to permeabilise the outer mitochondrial membrane, releasing cytochrome *c* and activating caspase-9-dependent intrinsic apoptosis [5]. The resulting injury cascade — from platinum adduct formation through NER engagement, oxidative amplification, and apoptotic execution — depends on multiple molecular processes, several of which show circadian oscillation; the extent of normal-tissue injury produced by a given dose therefore potentially depends on *when* that dose is administered.

Section 4 examines how these mechanisms manifest as organ-specific toxicities before Section 5 returns to the question of circadian timing.

3. Cisplatin entry points in OSCC/HNSCC treatment pathways: staging, treatment intent, practice patterns, and timing constraints

Oral squamous cell carcinoma (OSCC) and the broader group of head and neck squamous cell carcinomas (HNSCC) encompassing oropharynx, hypopharynx, and larynx primaries share a treatment framework anchored in TNM staging and the determination of surgical resectability. Cisplatin enters this framework at multiple decision nodes: as concurrent systemic sensitisation during radiation therapy in both postoperative and definitive settings, and as the backbone of induction chemotherapy in selected high-burden cases.

3.1. Surgery-first paradigm and the postoperative cisplatin entry point

For resectable OSCC and HNSCC, surgery remains the definitive treatment for the primary tumour and regional lymph nodes. Locoregional recurrence rates following surgery alone were historically 30–40% in stage III–IV disease, driving evaluation of adjuvant strategies. Two landmark phase III randomised trials — the European Organisation for Research and Treatment of Cancer (EORTC trial #22931; [3]) and the Radiation Therapy Oncology Group (RTOG trial #9501; [10]) — independently demonstrated that cisplatin at 100 mg m⁻² on days 1, 22, and 43 concurrent with postoperative radiotherapy (60–66 Gy) improved locoregional control and disease-free survival compared with radiotherapy alone. A joint comparative analysis of the two trials identified two histopathological features as the primary determinants of benefit: microscopically involved resection margins and extracapsular extension (ECE) of nodal disease. In the subset of patients with one or both of these high-risk features, pooled analysis demonstrated a 42% reduction in the

risk of locoregional treatment failure (HR 0.546) [11]. Patients without either risk factor derived no statistically significant survival advantage from the addition of chemotherapy, establishing a risk-stratified framework for postoperative cisplatin use that remains current guideline practice [11].

125 3.2. Definitive chemoradiotherapy and organ-preservation settings

For locally advanced HNSCC judged unresectable, or where organ preservation is the stated intent, concurrent cisplatin-based chemoradiotherapy is the first-line approach. Adelstein and colleagues established the superiority of cisplatin-based definitive concurrent chemoradiotherapy over radiotherapy alone for unresectable disease in a three-arm phase
130 III trial ([12]). The RTOG 91-11 trial confirmed concurrent cisplatin as the preferred systemic component for larynx preservation compared with sequential chemotherapy–radiotherapy or radiotherapy alone ([13]). For patients with very high nodal or tumour burden where induction chemotherapy is considered, the TAX 324 and TAX 323 phase III trials established TPF (docetaxel, cisplatin, fluorouracil) induction as superior to PF
135 (cisplatin, fluorouracil) alone ([14]; [15]). Across all concurrent settings, the MACH-NC collaborative group meta-analysis quantified the population benefit: adding concomitant platinum-based chemotherapy to radiotherapy conferred an absolute overall survival gain of approximately 6.5 percentage points at 5 years, with a consistent effect across oropharyngeal, oral cavity, laryngeal, and hypopharyngeal primaries ([16]).

140 The development of cetuximab, an anti-EGFR monoclonal antibody, introduced the first serious challenge to cisplatin’s position as the concurrent radiosensitiser of choice. Bonner and colleagues demonstrated that cetuximab plus radiotherapy significantly improved locoregional control and overall survival versus radiotherapy alone in locally advanced HNSCC ([17]), establishing cetuximab as an alternative concurrent agent. However, two
145 subsequent phase III trials have firmly restored cisplatin as the preferred partner for radiotherapy in the HPV-positive oropharyngeal subgroup that might otherwise be considered for de-intensification. The NRG RTOG 1016 trial demonstrated that radiotherapy plus cetuximab was inferior to radiotherapy plus cisplatin for overall survival in HPV-positive oropharyngeal cancer ([18]), and the contemporaneous De-ESCALaTE HPV trial reached
150 the same conclusion, showing significantly lower overall and progression-free survival in the cetuximab arm ([19]). Taken together, these trials confirm that cisplatin remains the

mandatory concurrent radiosensitiser in HPV-positive oropharyngeal cancer at standard risk and re-established the cisplatin backbone across HNSCC.

The argument for incorporating checkpoint immunotherapy into concurrent chemoradiotherapy was tested in two large phase III trials, both of which failed to meet their primary endpoints. Neither KEYNOTE-412 (pembrolizumab plus cisplatin-based CRT) ([20]) nor JAVELIN Head and Neck 100 (avelumab plus cisplatin-based CRT) ([21]) improved event-free or progression-free survival over cisplatin-based CRT alone. These negative results confirm that cisplatin-based concurrent chemoradiotherapy remains the standard of care against which toxicity reduction strategies, including chronotherapy, must be evaluated.

3.3. Weekly versus three-weekly scheduling: an unresolved debate with direct relevance to toxicity

The cisplatin dose schedule that established the evidence base described above — 100 mg m⁻² intravenously on days 1, 22, and 43 concurrent with radiotherapy — remains the regulatory and guideline standard. However, this regimen generates substantial acute toxicity: up to 40% of patients in the landmark postoperative trials could not complete all three planned high-dose cycles [3, 4]. Consequently, lower-dose weekly cisplatin schedules (typically 30–40 mg m⁻² per week for six or seven cycles) entered widespread institutional practice on the assumption of comparable efficacy with reduced toxicity [22].

The evidence for schedule equivalence has been vigorously contested. A systematic review and meta-analysis of 52 prospective studies comprising 4,209 patients found no overall survival difference between weekly and three-weekly cisplatin regimens in either the postoperative or the definitive setting; the weekly approach was significantly less nephrotoxic and myelosuppressive in the definitive setting, but was associated with more dysphagia and weight loss in the postoperative setting [22]. The Japanese JCOG1008 phase III randomised trial found weekly cisplatin at 40 mg m⁻² non-inferior to three-weekly 100 mg m⁻² for overall survival in postoperative high-risk HNSCC (HR 0.69; 99.1% CI 0.37–1.27, within the pre-specified non-inferiority margin of HR ≤ 1.54) with less grade ≥3 neutropenia and renal impairment in the weekly arm [23]. By contrast, a phase III randomised trial from Tata Memorial Hospital comparing weekly 30 mg m⁻² with three-weekly 100 mg m⁻² in a predominantly oral cavity adjuvant cohort found three-weekly cisplatin superior for

2-year locoregional control (73.1% vs 58.5%; HR 1.76; $p = 0.014$) [24], demonstrating that weekly cisplatin at 30 mg m^{-2} does not reliably substitute for the standard schedule in all OSCC contexts. Importantly, the cumulative cisplatin dose — regardless of schedule — is a dose-limiting toxicity predictor: a systematic review identified 200 mg m^{-2} as the threshold above which nephrotoxicity and ototoxicity rates rise markedly [4], providing an additional scheduling constraint that is independent of and complementary to circadian timing. The divergent results of these trials underline that toxicity management, not schedule optimisation alone, remains a first-order clinical problem — and that weekly schedules, by providing six to seven potential timing-intervention windows rather than three, may actually offer an advantage for chronotherapy implementation.

3.4. International guideline convergence and regional practice variation

International guidance converges on a shared therapeutic backbone while leaving considerable room for regional adaptation. In the United States, the NCCN framework treats cisplatin-based concurrent chemoradiotherapy as the reference systemic partner for definitive or postoperative radiotherapy in fit patients with locoregionally advanced HNSCC, a position grounded in the postoperative EORTC 22931 and RTOG 9501 trials and reinforced by later data showing inferior outcomes when cetuximab replaces cisplatin in cisplatin-eligible HPV-positive disease [25, 3, 10, 18, 19]. The ASCO neck-management guideline is similarly surgery-centred for oral cavity cancer, emphasising adequate neck management and selective use of postoperative radiotherapy or chemoradiotherapy when pathological risk factors justify treatment intensification [26].

European and UK recommendations frame the same principle through a more explicitly multidisciplinary oral-cavity pathway. The EHNS–ESMO–ESTRO guideline covers squamous carcinomas of the oral cavity, larynx, oropharynx, and hypopharynx and endorses surgery followed by risk-adapted radiotherapy or chemoradiotherapy for resectable oral cavity disease [27]. The UK National Multidisciplinary Guideline is especially explicit for oral cavity and lip cancer: surgery remains the mainstay for oral cavity tumours, clinical clearance of approximately 1 cm is recommended when vital structures permit, elective neck treatment is recommended for oral cavity tumours, and adjuvant radiochemotherapy improves control in the presence of advanced neck disease or positive margins [28]. Importantly for the present review, the same UK guideline recognises cisplatin 100 mg m^{-2}

every three weeks as the best-known concurrent regimen while acknowledging that other
215 platinum schedules are used in practice, capturing the gap between guideline reference
regimens and real-world deliverability [28].

Asian practice illustrates why this gap matters. The Pan-Asian adaptation of the
EHNS–ESMO–ESTRO guideline was developed specifically to translate the European
recommendations into Asian health-care settings, including expert representation from
220 Japan, India, Taiwan, Korea, China, Malaysia, and Singapore [29]. In Japan, JCOG1008
provides randomised evidence supporting weekly cisplatin 40 mg m^{-2} as a non-inferior
postoperative option when adequate cumulative exposure is maintained [23]. In India,
the Tata Memorial trial provides the important counterpoint that weekly cisplatin at
 30 mg m^{-2} was inferior to three-weekly high-dose cisplatin for locoregional control in a
225 predominantly oral cavity adjuvant population [24]. These two trials are not contradictory
so much as dose- and context-specific: they show that “weekly cisplatin” is not a single
intervention, and that both dose per infusion and cumulative delivered dose must be
specified.

Taiwan is a particularly relevant real-world context because of the high burden of
230 oral cavity squamous cell carcinoma associated with betel-quid exposure. Taiwanese
practice generally follows the same cisplatin-based chemoradiotherapy backbone, but
treatment is frequently adapted for renal function, nutritional compromise, mucositis
risk, and performance status. Population-based Taiwanese data in stage III–IV HNSCC
associated total cisplatin dose $\geq 240\text{ mg m}^{-2}$ with survival-related prognostic significance,
235 while a postoperative high-risk oral cavity cohort found that a larger proportion of patients
receiving three-weekly cisplatin achieved the commonly cited $\geq 200\text{ mg m}^{-2}$ cumulative
threshold than those receiving weekly low-dose cisplatin [30, 31]. For Taiwan and other
high-OSCC-burden settings, the practical problem is therefore not whether cisplatin is the
standard backbone, but how to preserve adequate cumulative exposure while reducing the
240 organ-specific toxicities that drive dose delay, dose omission, and treatment interruption.
This is precisely the clinical niche in which chronotype-aware scheduling could be tested: as
a toxicity-reduction strategy layered onto existing regional standards, not as a replacement
for surgery, radiotherapy, or cisplatin itself.

3.5. *Delivery logistics and the chronobiological blind spot*

245 In practice, cisplatin administration within a chemoradiotherapy programme is constrained by clinically rational parameters: creatinine clearance threshold, baseline and on-treatment audiometry, haematological reserve, and the logistics of intravenous infusion with hyperhydration and antiemetic prophylaxis. For the three-weekly regimen, the three infusions on days 1, 22, and 43 of the radiation course are determined by the treatment
250 calendar; for weekly schedules, timing aligns with the radiotherapy fraction timetable of the treating institution.

None of these constraints incorporate any consideration of when, within the 24-hour biological cycle, a patient is best positioned to receive cisplatin. The NER capacity of the patient's normal tissues is not assessed at the time of scheduling. The circadian
255 peak of OCT2 expression in the renal tubule and cochlea is not documented. The hour of cisplatin infusion is determined by ward schedule, infusion-centre capacity, and radiotherapy appointment time — not by patient biology. Current cisplatin protocols are operationally precise but chronobiologically silent.

4. **Why cisplatin injures specific organs: tissue uptake, DNA damage, oxidative stress, and repair failure**

260

Although cisplatin distributes systemically, its toxicities concentrate in a small number of tissues that share three features: disproportionately high platinum uptake via specific transporters, elevated metabolic demand that amplifies oxidative injury, and limited regenerative capacity once a damage threshold is exceeded. The kidney proximal tubule,
265 cochlear outer hair cells, oral mucosa, dorsal root ganglion neurons, and the upper gastrointestinal tract each exemplify a different combination of these vulnerabilities. Table 1 summarises the principal injury mechanisms and potential circadian timing gates for each toxicity category.

4.1. *Nephrotoxicity*

270 Nephrotoxicity is the dose-limiting toxicity of cisplatin [32]. Acute kidney injury (AKI) occurs in approximately 20–30% of treated patients under contemporary hydration protocols; before the introduction of forced saline diuresis, the incidence approached

100% [32]. Hypomagnesaemia, reflecting injury to tubular resorption machinery, affects 40–100% of patients [32].

275 The kidney concentrates cisplatin substantially above plasma levels because CTR1 and OCT2 are co-expressed at the basolateral membrane of proximal tubular cells. Genetic deletion of OCT2 in mice produced an approximately four-fold reduction in renal platinum accumulation and conferred near-complete protection from clinically significant nephrotoxicity [32]. The S3 segment of the proximal tubule is the most vulnerable
280 compartment, partly because its exceptional mitochondrial density creates an environment of intense oxidative metabolism [32, 2]. A pharmacokinetically important observation is that only approximately 1% of intracellular cisplatin reaches the nucleus; lanthanide-based immunoassays consistently detect four- to six-fold higher platinum adduct burdens in mitochondrial than in nuclear DNA, identifying the mitochondrion as a disproportionate
285 cisplatin damage target [32].

Intracellular injury converges on multiple death-signalling cascades. DNA damage stabilises p53, which transactivates PUMA- α ; PUMA- α sequesters the antiapoptotic proteins Bcl-2 and Bcl-X_L, releasing BAX to permeabilise the outer mitochondrial membrane and initiate cytochrome *c* – caspase-9 apoptosis [5, 2]. In parallel, p53-induced
290 protein with a death domain (PIDD) activates caspase 2 and releases apoptosis-inducing factor (AIF) from mitochondria [2]. Reactive oxygen species generated by impaired mitochondrial respiration and cytochrome P450 microsomal activity further amplify this injury [2, 32]. Cisplatin additionally triggers an innate inflammatory response: damage-associated molecular patterns (DAMPs) engage Toll-like receptor 4, activating NF- κ B
295 and driving TNF- α -mediated cytokine and chemokine production that sustains tubular injury beyond the initial platinum insult [32].

4.2. Ototoxicity

Cisplatin causes permanent, predominantly high-frequency sensorineural hearing loss in 40–80% of adult patients and in at least 50% of children [33]. Cochlear outer hair cells of
300 the basal turn are the primary targets, but damage to the stria vascularis — the structure responsible for maintaining the endolymphatic potential — contributes substantially to auditory dysfunction [33, 2].

A pharmacokinetic property unique to the cochlea explains its exceptional vulnerability.

Using inductively coupled plasma mass spectrometry (ICP-MS), Breglio and colleagues
305 demonstrated that, whereas cisplatin is detected in most organs within one hour and
cleared over days to weeks, the cochlea retains platinum species for months to years after
the end of chemotherapy [33]. During a three-cycle clinical-model regimen, platinum
accumulated progressively in the cochlea with each cycle without intercycle clearance [33].
At 60 days after the final cycle, serum platinum had fallen by 89%, yet platinum in
310 the stria vascularis had decreased by only approximately 50%, the organ of Corti by
approximately 39%, and the spiral ganglion by approximately 25% [33]. Postmortem
human cochlear sections retained detectable platinum at least 18 months after the final
cisplatin cycle [33]. Endocochlear potentials were significantly reduced after the first
treatment cycle and remained impaired 60 days post-regimen, demonstrating that stria
315 vascularis dysfunction precedes and outlasts acute hair-cell loss [33].

The molecular injury cascade within the cochlea relies heavily on oxidative rather than
direct DNA-adduct cytotoxicity: cisplatin inhibits glutathione peroxidase, glutathione S-
transferase, and superoxide dismutase, shifting the cochlear redox state; lipid peroxidation
generates 4-hydroxynonenal (4-HNE), which induces pathological Ca^{2+} influx in outer
320 hair cells and triggers mitochondria-dependent apoptosis [2]. Hearing loss is permanent
because post-natal cochlear outer hair cells lack regenerative capacity.

Otoprotective strategies are under active clinical development. The COG ACCL0431
phase III trial established that sodium thiosulfate (STS), administered 6 hours post-
cisplatin, significantly reduced hearing loss in paediatric patients receiving cisplatin-based
325 therapy compared with observation [34]. The SIOPEL 6 trial subsequently confirmed that
STS otoprotection was achievable without compromising antitumour efficacy specifically
in children with hepatoblastoma [35], supporting FDA approval of STS (Pedmark) for
this indication in September 2022. However, direct extrapolation to adult HNSCC raises
a substantive concern: STS binds and inactivates extracellular cisplatin, and preclinical
330 data suggest that systemic STS may reduce tumour cisplatin exposure. In adult HNSCC
— where cisplatin serves as a radiosensitiser against locoregional disease rather than a
solely systemic agent — the risk of antitumour antagonism has substantially limited STS
adoption and explains why the cochlear platinum retention documented by Breglio et al.
remains an unresolved clinical problem in this population [33].

Oral mucositis is the toxicity most directly visible in the OSCC/HNSCC treatment field. It occurs in approximately 20–40% of patients receiving conventional chemotherapy and in nearly all patients undergoing head and neck radiation therapy [36]. Its clinical impact is disproportionate: pain requiring parenteral opioids, dysphagia mandating enteral
 340 nutrition, and mucositis-driven treatment interruptions are consistent predictors of inferior treatment completion and adverse outcomes in HNSCC [37, 36].

Sonis proposed a five-stage mechanistic model that remains the field’s organising framework [37]. In the *initiation* stage, chemotherapy and radiation produce DNA strand breaks in basal epithelial cells and submucosal fibroblasts while reactive oxygen species
 345 initiate additional non-DNA injury. The *primary damage response* stage is characterised by p53 activation and robust NF- κ B induction; NF- κ B can upregulate up to 200 downstream genes, including those encoding TNF- α , IL-1 β , and IL-6 [37]. During *signal amplification*, these cytokines activate the MAPK, JNK, and COX-2 pathways; TNF- α induces matrix metalloproteinases MMP1 and MMP3 that degrade the submucosal extracellular matrix,
 350 and ceramide generated through sphingomyelinase activation promotes fibroblast and endothelial apoptosis [37]. The *ulcerative* stage, developing approximately 10 days after a stomatotoxic chemotherapy cycle, is the most clinically significant: epithelial integrity fails, bacteria colonise the ulcer bed, and in neutropenic patients the portal bacteraemia represents a life-threatening complication [37]. *Healing* occurs 2–3 weeks after cessation
 355 of the offending therapy, driven by extracellular matrix signals governing epithelial cell migration and differentiation [37].

4.4. Nausea and vomiting

Cisplatin is classified as a highly emetogenic chemotherapy agent [38]. Acute emesis (≤ 24 hours) is driven by cisplatin-induced serotonin release from intestinal enterochromaffin
 360 cells, activating vagal 5-HT₃ receptors and relaying the emetic signal to the nucleus tractus solitarius; delayed emesis (> 24 hours) is mediated predominantly by the NK₁/substance P pathway acting at the area postrema [38]. Contemporary prophylaxis combining a 5-HT₃ receptor antagonist, an NK₁ receptor antagonist, and dexamethasone substantially controls acute emesis, but delayed nausea remains incompletely suppressed in a significant
 365 proportion of patients [38, 39].

4.5. *Peripheral neuropathy*

Cisplatin-induced peripheral neuropathy (CIPN) is predominantly sensory, manifesting as a length-dependent, symmetrical “glove-and-stocking” distribution of paraesthesias, dysaesthesias, and numbness [40]. Cumulative dose is the primary risk determinant, with clinically significant neuropathy typically emerging above approximately 350 mg m⁻² cumulative [40]. Platinum compounds preferentially accumulate in dorsal root ganglion (DRG) neurons, which are particularly vulnerable because the blood–nerve barrier provides substantially less protection at the DRG than at peripheral nerve trunks; platinum–DNA adduct formation in sensory neuron nuclei drives cell death [40]. Mitochondrial dysfunction contributes an additional axonal injury component. A clinically important feature is the “coasting” phenomenon: neuropathic symptoms may continue to worsen for months after the final cisplatin dose as the full extent of DRG damage becomes manifest [40]. Persistent CIPN affects up to 40% of cancer survivors following platinum-based treatment [40]; however, in HNSCC practice the typical cumulative cisplatin dose (200–300 mg m⁻²) remains below the classical neuropathy threshold of approximately 350 mg m⁻² established in testicular and ovarian cancer series [4], making severe CIPN less prevalent in this population than nephro-, oto-, or mucosa-toxicity.

4.6. *Myelosuppression*

Cisplatin monotherapy produces modest myelosuppression relative to alkylating agents. In HNSCC chemoradiotherapy practice, however, cisplatin is frequently combined with fluorouracil, a taxane, or cetuximab, and cumulative haematopoietic progenitor injury contributes to dose reductions, treatment delays, and infection risk in these combination settings [22].

4.7. *Converging mechanisms and the rationale for timing*

Across all of these toxicity domains, a unifying pattern emerges: cisplatin injury severity is determined not merely by dose, but by the tissue-specific intersection of platinum uptake capacity, DNA adduct accumulation relative to repair competence, reactive oxygen species load relative to antioxidant buffering, and the regenerative reserve of the affected cell population. Each of these determinants is physiologically regulated and therefore potentially time-dependent. The circadian clock exerts documented control over

transporter expression, nucleotide excision repair activity, glutathione redox cycling, and mucosal epithelial proliferation — the molecular processes that appear, organ by organ, in the injury mechanisms described above. Section 5 examines the evidence that these processes oscillate with a periodicity clinically relevant to cisplatin administration.

5. Circadian timing gates in cisplatin uptake, repair, redox control, and tissue vulnerability

The molecular clock mechanism — a transcription–translation feedback loop anchored by the CLOCK/BMAL1 heterodimer and repressed by Period (PER) and Cryptochrome (CRY) proteins — does not merely govern sleep–wake cycles and metabolism. It imposes temporal structure on at least five processes that directly determine how much cisplatin-induced injury accumulates in normal tissues: nucleotide excision repair capacity; renal transporter expression; glutathione redox buffering; mucosal epithelial proliferation; and, in a paradoxical fashion, the apoptotic and inflammatory responses themselves. Each represents a potential gate through which the circadian time of administration could modulate toxicity independently of dose. Table 2 summarises all six gates with their evidence base. A human tissue clock atlas has catalogued the rhythmic expression of thousands of genes across 13 tissues [41], providing genomic-level support for the oscillatory framework reviewed here, while clinical cancer chronobiology reviews confirm its relevance across tumour types [42, 43].

Evidence gap: It is important to note that the mechanistic timing gates described below are characterised primarily in rodent tissues and cell lines. Direct evidence that XPA, OCT2, or glutathione oscillate with clinically meaningful amplitude in the kidneys, cochlea, or oral mucosa of OSCC or HNSCC *patients* is absent. Table 2 identifies the evidence strength and tissue-specific gaps for each gate. The mechanistic case for chronotherapy is therefore *plausible* rather than proven, and the proposed trial framework (Section 7) is designed specifically to test whether these preclinical timing effects translate to measurable human outcomes.

5.1. XPA and the circadian control of nucleotide excision repair

XPA is the DNA-damage verification scaffold of the NER pathway — the rate-limiting step at which the repair machinery confirms the presence of a helix-distorting lesion and

Table 1: Organ-specific mechanisms of cisplatin toxicity and potential circadian timing gates

Toxicity	Target tissue	Access route	Injury mechanisms	Potential timing gate	Clinical relevance
Nephrotoxicity	Proximal tubular cells (S3 segment)	OCT2, CTR1	DNA damage, ROS, mitochondrial injury, p53–PUMA/BAX apoptosis, inflammation, tubular biomarker elevation	OCT2-mediated uptake, XPA/NER repair, BMAL1-linked renal injury	Dose-limiting; GFR decline, treatment delay
Ototoxicity	Cochlear outer hair cells, stria vascularis	OCT2-related accumulation	ROS, mitochondrial dysfunction, irreversible hair-cell apoptosis	OCT2 cochlear uptake, oxidative stress rhythms	Permanent hearing loss; survivorship
Mucositis	Oral/GI mucosal epithelium	Systemic exposure; RT synergy	DNA damage, epithelial-turnover failure, NF- κ B amplification, barrier breakdown	Mucosal proliferation rhythm, epithelial repair timing, clock-controlled cell cycle	Pain, dysphagia, opioids, feeding tube, treatment interruption
Nausea / vomiting	GI enterochromaffin cells; central emetic pathways	Systemic exposure	Serotonin release, vagal afferents, NK1/substance P pathway	GI motility and neuroendocrine rhythms	Acute toxicity; hydration, nutrition impact
Neuropathy	Dorsal root ganglion neurons	Platinum accumulation; limited blood-nerve barrier	Mitochondrial dysfunction, ROS, axonal degeneration, sensory neuron apoptosis	Mitochondrial redox state, axonal repair capacity	Long-term sensory morbidity
Myelosuppression	Bone marrow progenitors	Systemic exposure	Progenitor injury	Hematopoietic progenitor cycling	Combination-regimen relevant

assembles the dual-incision complex around it. Its oscillating abundance is the best-characterised circadian gate for cisplatin toxicity. Kang and colleagues demonstrated that CLOCK/BMAL1 drives *Xpa* gene transcription directly, while CRY proteins — themselves products of clock-driven transcription — promote XPA proteasomal degradation via the HERC2 ubiquitin ligase [44]. The opposing drives produce a robust oscillation: XPA protein peaks at approximately Zeitgeber time (ZT) 10 and reaches its nadir at approximately ZT22, with an amplitude spanning five- to ten-fold in mouse brain, liver, and kidney [44]. In *Cry1/Cry2* double-knockout mice the oscillation is abolished; XPA is constitutively expressed at an intermediate level, confirming that CRY is indispensable for the trough [44]. Lee and Sancar extended this work, demonstrating that the XPA oscillation governs the rate at which NER removes cisplatin–DNA adducts in vivo: repair activity is fastest near ZT10 and slowest near ZT22 [45, 46]. Because unrepaired adducts drive tubular cell apoptosis and cochlear hair-cell death (Section 4), this rhythm implies that normal tissues are most vulnerable to cisplatin-induced killing in the late subjective night, when repair capacity is lowest.

5.2. *OCT2/SLC22A2 and rhythmic cisplatin influx into the kidney*

A complementary gate governs the quantity of cisplatin that enters proximal tubular cells. OCT2, encoded by *Slc22a2*, is the dominant basolateral uptake transporter for cisplatin in the kidney and the primary determinant of intratubular platinum accumulation (Section 4). Oda and colleagues established that the CLOCK transcription factor transcriptionally induces peroxisome proliferator-activated receptor α (PPAR α), which in turn occupies the *Slc22a2* promoter and drives rhythmic OCT2 protein accumulation [47]. In mice, renal OCT2 protein peaks near ZT14. Consistent with this, a single cisplatin dose administered at ZT6 — rising OCT2, highest tubular influx — produced significantly greater nephrotoxicity than the same dose at ZT18 [47]. In *Clock* mutant mice lacking CLOCK transcriptional activity, OCT2 oscillation was abolished and the dosing-time difference in nephrotoxicity disappeared entirely, establishing a causal rather than merely correlative dependence of the timing effect on an intact molecular clock [47]. The OCT2 gate is thus the most mechanistically complete circadian determinant of cisplatin nephrotoxicity characterised to date.

5.3. *Glutathione, redox rhythms, and the oxidative injury window*

Cisplatin forms conjugates with reduced glutathione (GSH), diverting reactive platinum away from DNA; GSH depletion amplifies ROS-mediated tubular and cochlear injury (Section 4). Hepatic GSH abundance shows a circadian rhythm in rodents, with troughs coinciding with the resting phase [48]. Although the tissue-specific GSH oscillation in kidney and oral mucosa has not been characterised at the resolution of the XPA and OCT2 systems, rhythmic redox capacity is a well-established principle of clock-controlled metabolism. A window of diminished GSH would be expected to compound the adduct accumulation already predicted by low NER activity, creating a toxicity risk that is driven by converging vulnerability rather than any single mechanism.

5.4. *Mucosal epithelial proliferation and the mucositis timing window*

Bjarnason and colleagues performed the only direct molecular-clock measurement in human oral mucosal tissue, sampling biopsies at multiple time points across 24 hours from healthy volunteers [49]. Clock gene expression was robustly rhythmic: hPer1 peaked at approximately 07:58, hCry1 at 17:04, and hBmal1 at 21:40 — confirming that the autonomous circadian oscillator operates in normal oral epithelium with the expected phase relationships. Critically, thymidylate synthase (TS) activity — a functional marker of S-phase proliferative commitment — peaked at approximately 13:27, identifying a mid-afternoon window of maximum mitotic activity [49]. Because cisplatin-induced DNA damage is most cytotoxic to actively dividing cells, a DNA-damaging insult delivered at the proliferative peak would be predicted to cause more extensive mucosal injury than the same dose given during the quiescent evening window. An additional layer of complexity is introduced by the gut microbiome, whose composition and metabolic activity oscillate diurnally in a clock-dependent manner [50]; microbial circadian rhythms influence mucosal barrier function and systemic inflammatory tone in ways that may modulate mucositis severity, though their specific relevance to cisplatin-induced mucosal injury in HNSCC has not been characterised.

5.5. *The BMAL1/ARNTL tissue-specific paradox*

Among the clock components with direct evidence in head-and-neck oncology, BMAL1 (ARNTL) presents the most instructive paradox. Three independent

lines of evidence assign it opposing roles depending on tissue context.

First, in a cisplatin-treated mouse model, Zha and colleagues found that *Bmal1* knock-out significantly reduced cisplatin-induced acute kidney injury: the wild-type BMAL1-competent kidney was more susceptible to tubular cell death, suggesting that BMAL1 facilitates — rather than protects against — renal cisplatin toxicity, possibly by coordinating pro-inflammatory NF- κ B and pro-apoptotic signalling in tubular cells [51]. Second, in OSCC, Li and colleagues demonstrated that BMAL1 acts as a tumour suppressor: high BMAL1 expression correlates with better prognosis, and BMAL1 promotes autophagy-mediated death in OSCC cells; cisplatin-induced BMAL1 upregulation may therefore enhance tumour sensitivity [52]. Third, in nasopharyngeal carcinoma (NPC) — a closely related SCC of the head and neck — Peng and colleagues showed that epigenetic silencing of *ARNTL* (BMAL1) promotes cisplatin resistance, while restoration of BMAL1 expression resensitises cells to cisplatin [53].

Taken together, these three lines of evidence reveal a two-tissue asymmetry with an important internal structure. On the *tumour side*, Li 2022 and Peng 2019 are concordant: BMAL1 promotes cisplatin sensitivity in both OSCC and NPC. On the *host side*, Zha 2020 suggests that BMAL1 facilitates renal tubular injury, but this finding comes from a single laboratory and has not been independently replicated; it should be treated as a hypothesis-generating signal rather than a settled mechanism. The framing that best fits the current data is therefore one of a *host–tumour clock contrast*: clock-active tumour cells are more cisplatin-sensitive, while the renal host response to clock activity may add to toxicity. If confirmed, this asymmetry would be exploitable — a timing window that reduces renal OCT2-driven uptake while maintaining tumour BMAL1 activity and cisplatin sensitivity. A naive framework in which “restoring or amplifying the clock reduces toxicity” remains insufficient because it conflates host and tumour clock function in opposing biological contexts.

5.6. *Host–tumour clock asymmetry as a positive therapeutic mechanism*

The BMAL1 data reviewed above suggest a reframing that moves beyond the defensive question of “how do clock-disrupted patients lose the timing benefit?” and towards the positive question of “does the host–tumour clock contrast itself constitute the therapeutic window?” HNSCC tumours frequently exhibit attenuated or phase-shifted circadian oscil-

lations, driven by hypoxia, metabolic reprogramming, and disrupted cell-cycle control [54]. If OSCC tumour cells have low BMAL1 activity and thus diminished circadian clock amplitude, they would fail to upregulate the clock-driven cisplatin-repair machinery (XPA) during the subjective morning — the very window when host normal tissues are best positioned to repair cisplatin adducts. In this model, the host–tumour clock contrast is not a complication to manage but the mechanistic basis of the therapeutic window: normal tissues repair efficiently at the XPA/NER peak, tumour cells cannot. The *host–tumour clock contrast* — the difference in circadian amplitude between normal and tumour cells — could itself be a biomarker of likely chronotherapy benefit. This reframing has a direct translational implication: the exploratory biomarker objectives in Section 7 should include tumour BMAL1/PER2 IHC as a predictor of differential benefit, not only as a mechanistic correlate.

5.7. *PER/CRY proteins and the circadian integrity prerequisite*

PER and CRY proteins constitute the negative feedback limb of the core oscillator. Their roles extend beyond autoregulation: as shown in Section 2, CRY is directly responsible for HERC2-mediated XPA degradation, linking clock amplitude to NER capacity in a mechanistically direct manner [44]. Bjarnason’s oral mucosa data confirm that PER expression is rhythmic and phase-coherent in normal epithelium [49]. This represents a prerequisite for the timing-gate model: the expected benefit of time-of-delivery optimisation depends on a sufficiently intact circadian oscillation in host normal tissues, which cannot be assumed in the heavily treated OSCC/HNSCC patient population. A further implication is that the circadian function index (CFI) — measured from wrist actigraphy as proposed in Section 7 — captures *system-level* circadian integrity across all tissues, including those most relevant to cisplatin toxicity.

Collectively, the XPA/NER, OCT2, GSH, and mucosal-proliferation rhythms provide a multi-pathway mechanistic substrate through which the circadian time of cisplatin administration could shift the balance between normal-tissue toxicity and tumour kill. The BMAL1 host–tumour asymmetry underscores that this relationship is not unidirectional, and suggests that the host–tumour clock contrast may itself be a targetable feature rather than merely a confound. Whether these mechanistic predictions translate into clinically measurable toxicity differences is examined in Section 6.

Table 2: Circadian timing gates relevant to cisplatin toxicity and response

Gate / molecule	Function	Circadian evidence	Tissue relevance	Timing implication	Evidence strength
XPA / NER	Adduct verification; repair scaffold	Robust 10-fold oscillation in brain; in-phase with BMAL1	Kidney, liver, blood; OSCC mucosa/tumour unknown	Higher normal-tissue repair may reduce toxicity	Strong mechanism; HNSCC-specific gap
OCT2 / SLC22A2	Cisplatin uptake in kidney and cochlea	Clock-linked regulation suggested	Kidney, cochlea	Lower uptake phase may reduce nephro/ototoxicity	Strong transporter biology; timing less complete
CTR1 / SLC31A1	Cisplatin influx in tumour and normal cells	Circadian link unclear	Tumour cells, normal tissues	Important for efficacy; timing unestablished	Low-moderate
GSH / GCLC	Detoxification of reactive platinum species	Rhythmic redox regulation in some systems	Kidney, mucosa, tumour	May reduce normal-tissue injury; also reduces tumour kill	Hypothesis-generating
Mucosal cell-cycle regulators	Epithelial vulnerability and repair rate	GI/mucosal proliferation is rhythmic	Oral/GI mucosa	Low-proliferation or high-repair windows may reduce mucositis	Moderate
BMAL1 / ARNTL	Core clock factor; tissue-specific effects	Two-tissue asymmetry: host renal (Zha 2020 — single lab, unconfirmed); tumour-side concordance — OSCC tumour suppressor (Li 2022) and NPC cisplatin sensitiser (Peng 2019)	Kidney (host), OSCC/NPC tumour	Host-tumour clock contrast may be exploitable; Zha 2020 awaits independent replication	Moderate (tumour side); preliminary (host side)
PER / CRY	Negative clock limb; maintains circadian amplitude	Disruption abolishes chronotherapy benefit in preclinical models	Host tissues, immune system	Intact circadian rhythm may be a prerequisite for timing interventions	Strong preclinical
p53-PUMA/BAX	Apoptosis after unrepaired DNA damage	DNA damage response may be clock-modulated	Kidney, mucosa, tumour	Timing may shift apoptotic threshold	Mechanistically plausible
Inflammatory mediators	Amplify organ injury	Immune/inflammatory rhythms well documented	Kidney, mucosa	Timing may modulate inflammatory toxicity	Emerging
Gut microbiome (circadian oscillation)	Mucosal barrier; systemic inflammatory tone	Diurnal oscillation in composition and metabolism [50]	Oral/GI mucosa	May modulate mucositis; role in HNSCC uncharacterised	Hypothesis-generating

6. Clinical and preclinical evidence: promising toxicity signals but no proven survival benefit

550 The mechanistic case for cisplatin chronotherapy rests on the molecular evidence reviewed in Section 5; the clinical evidence base is promising but small. Table 3 summarises the key studies. The principal conclusion from this literature is that toxicity reduction is the most defensible near-term endpoint — survival benefit has not been demonstrated and is not a realistic primary endpoint for phase II study in this field.

555 6.1. OSCC pilot: dosing-time effects on toxicity and tumour response

Tsuchiya and colleagues reported a study in OSCC patients receiving 5-fluorouracil and cisplatin (DCF) induction chemotherapy that directly examined whether the timing of cisplatin administration affected the therapeutic and toxic outcome in an oral-cancer-specific context [55]. The clinical component involved a small crossover cohort ($n = 9$),
560 limiting statistical inference, but the study uniquely paired patient data with a parallel OSCC xenograft mouse model in which the same chemotherapy regimen was delivered at different Zeitgeber times. The xenograft experiments demonstrated significant dosing-time dependence in both antitumour efficacy and host toxicity [55], providing mechanistic corroboration in a tumour model that is directly relevant to the clinical question. The
565 human data suggested that evening cisplatin administration was associated with reduced severe nausea and vomiting relative to daytime delivery, consistent with the circadian antiemetic and mucosal pharmacology discussed in Section 4. Although the clinical sample is too small for efficacy conclusions, Tsuchiya et al. remains the only published study to examine cisplatin chronotherapy specifically in oral squamous cell carcinoma patients.

570 6.2. NPC phase II randomised trial: infusion-rate chronomodulation reduces acute toxicity

Zhang and colleagues conducted a phase II randomised study in patients with locally advanced nasopharyngeal carcinoma receiving concurrent cisplatin and radiotherapy [56]. The intervention warrants precise characterisation: patients were randomised to *chronomodulated infusion-rate* delivery — in which the rate of cisplatin infusion was varied according
575 to a pre-specified sinusoidal function targeting a fixed population-level low-toxicity phase — versus constant-rate infusion administered at the same clock time. The chronomodulated

arm showed reduced rates of clinically significant nausea, vomiting, and oral mucositis without evidence of compromised antitumour activity at the phase II sample size.

Two conceptual distinctions must be drawn before applying these results to the framework proposed in this review. First, Zhang 2018 tested *infusion-rate chronomodulation at a fixed clock time*: a pre-specified sinusoidal delivery profile applied uniformly to all patients, regardless of their individual circadian phase. This is not equivalent to *chronotype-anchored scheduling* — the personalised approach proposed in Section 7 — which calculates the delivery window individually from each patient’s measured circadian phase [57]. Zhang 2018 therefore provides mechanistic plausibility for the principle that the temporal pattern of cisplatin delivery within a dose modulates acute toxicity in a head-and-neck cancer context, but does not constitute direct evidence for the personalised chronotype-anchored strategy advocated here. Second, NPC differs from OSCC in its viral aetiology (EBV-driven) and biological behaviour; however, it shares the cisplatin-based concurrent chemoradiotherapy regimen and the oral/pharyngeal mucosal toxicity profile that is the primary target of chronotherapy. The mucositis and emesis findings are therefore cautiously transferable to the OSCC/HNSCC context, subject to the proviso that the study was not powered to detect survival differences.

6.3. Preclinical models: mechanistic support and translational constraints

Multiple independent mouse models support the principle of dosing-time-dependent cisplatin toxicity. The most directly relevant is Koritala et al., who demonstrated in a mouse mammary carcinoma model that cisplatin combined with ionising radiation produced different tumour growth inhibition and host toxicity profiles depending on the time of drug delivery [58]. The clock-dependent renal OCT2 oscillation established by Oda et al. [47] (Section 5) has been independently replicated and is consistent with broader evidence for rhythmic expression of renal efflux transporters including ABCC2 [59] and MDR1 [60], which modulate intratubular cisplatin retention. However, direct Zeitgeber-time-to-clock-time extrapolations from mouse to human are methodologically invalid: mice are nocturnal, so the ZT6 toxicity peak in Oda et al. corresponds to the onset of the active phase in a mouse, whereas the equivalent circadian phase in a diurnal human would fall in the early evening. Chronotype characterisation is therefore a prerequisite for translating preclinical timing findings to clinical practice.

6.4. Counter-evidence, sex interactions, and the circadian-disruption problem

The most sobering data from the chronotherapy literature come from the Giacchetti et al. EORTC phase III trial, the largest randomised study of platinum-based chronomodulated therapy ever conducted [61]. Among 564 patients with advanced colorectal cancer, chronomodulated oxaliplatin/5-fluorouracil/leucovorin improved objective response rate and reduced grade ≥ 3 mucositis and functional impairment versus conventional infusion — but this benefit was entirely concentrated in male patients. Women in the chronomodulated arm experienced *significantly worse* toxicity and inferior survival outcomes, a sex $\times$ timing interaction that was neither predicted by the preceding mechanistic framework nor identified in any preclinical model.

What the Giacchetti failure taught the field is threefold. First, the circadian phase relationships between molecular oscillators (XPA, OCT2, GSH) and sex-linked pharmacokinetics are not equivalent in women and men; a chronomodulated schedule optimised for average male chronobiology was actively harmful in women. Second, animal-to-human and sex-specific extrapolation cannot be assumed from rodent data and requires explicit experimental interrogation. Third, prospective sex-stratified analysis — with formal pre-specified interaction tests and independent safety monitoring of female participants — is a patient safety requirement in any future cisplatin chronomodulation trial, not a statistical option. Innominato and colleagues subsequently confirmed sex-related modulation of chronotherapy outcomes in the ARTDECO trial [62], establishing biological sex as a first-order independent variable in the circadian pharmacology of platinum-fluoropyrimidine regimens.

Seto et al. demonstrated in a non-small-cell lung cancer xenograft model that cisplatin dosing times that reduce nephrotoxicity do not necessarily reduce peripheral neuropathy in a parallel manner, raising the possibility of organ-specific timing trade-offs [63]. Finally, the circadian-disruption problem is fundamental: cancer patients frequently have blunted or phase-shifted rest–activity rhythms at baseline due to illness, sleep disruption, and the tumour microenvironment [64, 65], which would attenuate or eliminate the timing gates described in Section 5. Any clinical trial of cisplatin chronotherapy must therefore monitor circadian rhythm integrity as a stratification and biomarker variable, and must be sex-stratified by design.

The breadth of circadian pharmacology now extends to immune-checkpoint agents:
640 a multicentre analysis found that the time of immunotherapy infusion significantly affected overall survival in melanoma and lung cancer patients [66], indicating that chronosensitivity is not limited to cytotoxic platinum compounds and strengthening the general rationale for time-of-delivery research in oncology.

7. From fixed clock-time dosing to chronotype-aware cisplatin scheduling

645 The mechanistic and clinical evidence reviewed in Sections 5 and 6 supports the hypothesis that cisplatin toxicity in OSCC/HNSCC is modifiable by the circadian time of delivery, but translating this hypothesis into a trial-ready protocol requires five design commitments that have been absent from the handful of published studies: chronotype-aware rather than fixed-clock scheduling; toxicity as the primary endpoint rather than a secondary
650 measure; prospective circadian integrity assessment; biomarker integration for mechanistic interpretation; and systematic evaluation of workflow feasibility. Table 4 presents a schematic design for a proof-of-concept phase II randomised trial — *Chronotype-informed cisplatin scheduling for toxicity reduction in OSCC/HNSCC receiving chemoradiotherapy* — that embodies all five principles.

655 7.1. Design principle 1: chronotype-aware, not fixed clock

The principal failure mode of early chronotherapy trials was the application of a universal clock time — typically a fixed morning or evening infusion window — to an uncharacterised patient population. Circadian phase varies substantially between individuals: the midsleep-point on free days (the Munich Chronotype Questionnaire proxy for intrinsic circadian
660 phase) [57] varies by more than four hours across a healthy adult population and by greater amounts in elderly and oncology cohorts [65]. A fixed 6 a.m. cisplatin delivery coincides with the low-OCT2 window for evening chronotypes but with the high-OCT2 window for morning chronotypes, which would predict equal or increased nephrotoxicity. The proposed trial requires a 7-day pre-treatment actigraphy run-in to estimate individual
665 circadian phase, supplemented by the Munich Chronotype Questionnaire [57], and delivers cisplatin at a calculated time corresponding to approximately the mid-to-late subjective day.

Table 3: Evidence base for cisplatin chronotherapy relevant to OSCC/HNSCC

Study type	Model/population	Intervention	Main findings	Major limitation	OSCC/HNSCC relevance
OSCC crossover pilot	OSCC patients; DCF induction	Daytime vs evening cisplatin	Reduced severe nausea/vomiting	$n = 9$; no efficacy power	Direct but preliminary
NPC phase II RCT	Locally advanced NPC	Chronomodulated vs constant cisplatin	Reduced nausea, vomiting, mucositis	$NPC \neq OSCC$; not survival-powered	Supports mucosal/GI endpoint selection
Mouse cisplatin models	Tumour-bearing mice (non-HNSCC)	ZT-based cisplatin timing	Time-of-day affects toxicity and DNA repair	Cancer model mismatch	Mechanistic support
Clock-disruption models	Tumour-bearing jet-lagged mice	Normal vs disrupted rhythm	Disruption abolishes timing benefit	Preclinical only	Supports need for circadian monitoring
Population chronobiology cohorts	Cancer patients; rest-activity monitoring	Rest-activity rhythm assessment	Rhythm disruption linked to poorer outcomes	Not cisplatin-specific	Circadian integrity as prognostic biomarker

The delivery-time formula is: *Cisplatin infusion time = midsleep on free days (MSFsc) – 7 hours*. The –7 h offset is derived from the OCT2 oscillation: renal OCT2 peaks near ZT14 (~mid-subjective afternoon in diurnal organisms) and is at nadir near ZT2 (~early subjective morning) [47]. Dosing at MSFsc – 7 h targets a phase approximately 5–7 hours before the predicted OCT2 peak — the window during which OCT2-mediated platinum uptake is declining and XPA-driven NER capacity is approaching its zenith [44].

Worked examples for three chronotype strata:

- *Morning chronotype* (MSFsc 02:30): infusion at approximately 07:30 a.m.
- *Intermediate chronotype* (MSFsc 04:00): infusion at approximately 09:00 a.m.
- *Evening chronotype* (MSFsc 06:00): infusion at approximately 11:00 p.m. (the previous evening) or 11:00 a.m. if a phase-advance is applied for operational feasibility.

For the majority of patients (MSFsc between 02:00 and 05:00, consistent with adult population distributions [57]), chronotype-adjusted cisplatin would fall between approximately 07:00 a.m. and 10:00 a.m., overlapping with standard outpatient infusion hours. For late chronotypes (MSFsc > 05:00, approximately 10–15% of the adult population), the formula yields times outside standard hours, and an operational protocol for accommodating these patients is a pre-specified feasibility objective.

Phase-translation caveat: The –7 h offset is a pragmatic compromise primarily weighted toward the OCT2/nephrotoxicity gate, for which the clock-to-phase evidence is most complete [47]. XPA and OCT2 oscillate with a phase difference of approximately 4 hours (XPA peak ~ZT10 vs OCT2 peak ~ZT14); the proposed infusion time targets a position roughly midway between the OCT2 trough and the XPA zenith. These phase relationships may not be identical across all individuals, tissues, or clinical states, which is why continuous actigraphy-based phase monitoring throughout treatment is required rather than a single pre-treatment assessment.

7.2. Design principle 2: toxicity-first endpoints

Survival-focused endpoints are inappropriate for a proof-of-concept chronotherapy trial in OSCC/HNSCC. The proposed primary composite endpoint is the rate of Grade ≥ 3 mucositis (CTCAE v5.0) or Grade ≥ 3 nephrotoxicity within **six weeks of the first**

cisplatin infusion — a fixed calendar period chosen to capture the expected toxicity peak for both endpoints (mucositis peaks at weeks 4–6 of concurrent CRT; cumulative nephrotoxicity is most sensitive after two or more full-dose cycles). A calendar-period definition avoids the schedule-dependent ambiguity of “cycles 1–2”, which covers 6 weeks for a q3wk regimen but only 2 weeks for a weekly regimen [23]. Both endpoint components are directly plausible targets of the mechanistic pathways described in Sections 4 and 5, both drive treatment interruptions and dose reductions that compromise long-term outcomes [22], and both can be reliably assessed in a moderate-sized trial. The composite endpoint will be analysed as a hierarchy: if the composite endpoint is positive, mucositis and nephrotoxicity components will be tested sequentially under a pre-specified hierarchical testing procedure. A win-ratio analysis [67] will be performed as a pre-specified sensitivity analysis. Secondary toxicity endpoints include: emesis complete-response rate; audiometric hearing loss (SIOP Boston Scale); neuropathy severity (NCI CTCAE); cumulative cisplatin dose delivered; and the number of radiotherapy fractions missed due to toxicity. Patient-reported outcomes (EORTC QLQ-C30, H&N35) are co-primary from the patient perspective.

7.3. Design principle 3: circadian integrity assessment

Cancer patients frequently have disrupted rest–activity rhythms at baseline [64, 68], and treatment itself — inflammation, pain, anxiety, altered sleep hygiene during concurrent chemoradiotherapy — further attenuates circadian amplitude. The proposed trial incorporates continuous wrist actigraphy throughout treatment and employs the circadian function index (CFI) [69] and interdaily stability metrics to quantify rhythm disruption in each patient. A wearable skin temperature sensor, which tracks the distal–proximal skin temperature gradient as a non-invasive circadian phase marker [70], provides a complementary continuous signal. Patients will be stratified at enrolment into “intact rhythm” (CFI above threshold) versus “disrupted rhythm” strata. This stratification enables the trial to test the a priori hypothesis that the chronotherapy benefit is restricted to patients with measurable circadian amplitude, directly addressing the well-documented concern [64, 65] that timing interventions may be ineffective in clock-disrupted patients.

7.4. Design principle 4: biomarker integration

Pre-treatment peripheral blood leukocyte (PBL) clock gene expression (hPER1, hBMAL1) will be quantified at three time points across a circadian day (morning, afternoon, evening) to confirm the phase estimate from actigraphy and identify patients with severely flat molecular rhythms. This requires three timed blood draws within a single day, which for the majority of patients can be arranged on an outpatient basis; the protocol will specify whether inpatient or day-care attendance is required for patients whose individual chronotype makes same-day out-of-hours phlebotomy impractical.

Urinary platinum excretion will be collected at 2-hourly intervals during the first 8 hours post-infusion as a pharmacokinetic proxy for effective renal cisplatin clearance, which is predicted to be higher when OCT2 activity is lower. Serial collection requires a structured hydration and urine-collection protocol on the infusion day; institutional nursing capacity to implement this must be confirmed as part of the feasibility sub-study.

Tumour biopsy material from routine diagnostic FNAC or core biopsy will undergo BMAL1/PER2 immunohistochemistry (IHC) to explore the relationship between tumour clock status (Section 5) and treatment response. For BMAL1 IHC, a validated antibody clone (e.g., Abcam ab93806 anti-ARNTL, or equivalent with documented specificity in head and neck tissue) must be specified in the laboratory manual; staining will be scored using the H-score (0–300) with pathologist blinding to treatment allocation and outcome. For PER2 IHC, equivalent antibody validation and scoring criteria apply. Given the absence of a CAP/CLIA-validated companion diagnostic assay for circadian clock proteins, these biomarker analyses are designated as *exploratory only* and will not be used for treatment allocation or interim stopping decisions. Analytical validity must be confirmed with positive and negative tissue controls before study commencement.

7.5. Design principle 5: workflow feasibility

Chronotype-adjusted cisplatin delivery requires individualized infusion start times that may fall outside standard pharmacy preparation and nursing shift hours. The proposed trial therefore includes a feasibility sub-study in the first 20 patients, collecting data on: time from randomisation to actual infusion start; pharmacy preparation lead time required; nursing staff acceptability; and patient preference for scheduling. Cisplatin can be varied independently of radiotherapy timing, so no modification to the radiotherapy workflow

is required. Preliminary modelling suggests that for most patients, chronotype-adjusted cisplatin would fall between 10 a.m. and 6 p.m., overlapping substantially with standard outpatient infusion hours and limiting the logistic burden [71, 72].

760 7.6. Chronotherapy in the context of established toxicity-reduction strategies

Any evaluation of cisplatin chronotherapy must be contextualised within the established repertoire of toxicity-reduction strategies with which it would compete or complement in clinical practice. Several approaches have already demonstrated efficacy or are under active investigation.

765 *Radioprotection and hyperhydration:* Intensive intravenous hydration with forced saline diuresis and magnesium supplementation substantially reduces nephrotoxicity incidence and remains the current standard of care [32]. These manoeuvres address the *concentration* of cisplatin in the tubular lumen, whereas chronotherapy addresses the *transporter-mediated uptake* at the basolateral membrane — different mechanistic targets that are not mutually
770 exclusive.

Sodium thiosulfate (STS) otoprotection: As reviewed in Section 4, STS is approved for paediatric hepatoblastoma but carries antitumour antagonism concerns in adult HNSCC [35, 34]. Chronotherapy offers a mechanism-based alternative that avoids this trade-off.

775 *Amifostine:* The thiol radioprotector amifostine reduces nephrotoxicity and xerostomia but is associated with significant acute toxicity (nausea, hypotension, vomiting) that limits its tolerability in the chemoradiotherapy setting [22], and its routine use is not recommended. Chronotherapy would not be expected to interact with amifostine’s mechanism.

780 *Proton therapy, IMRT, and parotid-sparing radiotherapy:* These radiotherapy techniques reduce xerostomia and dysphagia but do not directly address cisplatin nephrotoxicity or ototoxicity — the principal dose-limiting toxicities. Chronotherapy is orthogonal to these radiotherapy-based strategies.

Schedule modification (weekly vs three-weekly): As reviewed in Section 3, schedule
785 modification alters the *cumulative dose rate* but not the *circadian timing* of any individual infusion. The two approaches are complementary: chronotherapy personalises *when* each dose is given; schedule modification determines *how much* and *how often*. Weekly schedules

Table 4: Schematic design: *Chronotype-informed cisplatin scheduling for toxicity reduction in OSCC/HNSCC receiving chemoradiotherapy*

Design element	Specification
Population	Histologically confirmed OSCC/HNSCC (stage III–IVA), planned for concurrent cisplatin (100 mg m^{-2} q3wk or 40 mg m^{-2} weekly) + RT; ECOG PS 0–1; no prior platinum; creatinine clearance $\geq 55 \text{ mL min}^{-1}$ (Cockcroft–Gault; preferred over eGFR due to muscle-mass confounding in cachectic HNSCC patients [4]); baseline audiogram required; no pre-existing SIOP Boston Grade ≥ 3 hearing loss; p16/HPV status documented at enrolment (stratification variable); pregnancy testing required for female patients of childbearing potential; reliable contraception for duration of treatment; ECOG PS2 permitted only with documented MDT agreement (increased toxicity and circadian disruption risk).
Chronotype assessment	7-day wrist actigraphy run-in + Munich Chronotype Questionnaire (MSFsc); skin temperature wearable from day 1 of RT. Cisplatin delivered at estimated MSFsc–7 h (individual-specific; see Section 7).
Comparator	Cisplatin at institutional fixed time (standard of care).
Primary endpoint	Rate of Grade ≥ 3 mucositis or Grade ≥ 3 nephrotoxicity within 6 weeks of first cisplatin infusion (fixed calendar window; CTCAE v5.0). Hierarchical testing of components if composite significant; win-ratio sensitivity analysis pre-specified.
Secondary endpoints	Emesis complete-response rate; SIOP Boston ototoxicity grade; neuropathy (CTCAE); cumulative cisplatin dose delivered; RT fractions missed; EORTC QLQ-C30/H&N35 (QoL); host–tumour clock contrast (tumour vs PBL BMAL1 expression ratio).
Stratification	Cisplatin schedule (q3wk vs weekly); tumour site (oral cavity vs oropharynx/hypopharynx/larynx); circadian integrity (intact vs disrupted CFI); sex; p16/HPV status (oropharynx patients).
Exploratory biomarkers	PBL clock gene expression (PER1, BMAL1; three time-points); serial urinary Pt excretion (2-hourly, first 8 h); tumour BMAL1/PER2 IHC (H-score; validated antibody; pathologist blinded); host–tumour clock contrast.
Sample size	Assumes 50% baseline composite Grade ≥ 3 rate (mucositis 35–45% [36] + nephrotoxicity 20–30% [32]; partial overlap; supported by concurrent CRT trial data). Sensitivity to absolute risk reduction (ARR): ARR 10%: $n = 180$ /arm; ARR 15%: $n = 110$ /arm; ARR 20%: $n = 80$ /arm (primary design target); ARR 25%: $n = 55$ /arm (all at 80% power, $\alpha = 0.10$ one-sided, χ^2 test with continuity correction). The primary design target of 80 per arm represents a minimum phase II signal-seeking design; the trial may be expanded to 110/arm if early interim data suggest the effect size is nearer 15%.
Statistical analysis plan	Primary: two-sided χ^2 with continuity correction; one-sided $\alpha = 0.10$. Pre-specified sex $\times$ timing interaction test (formal F -test; descriptive power only at 80/arm). D SMB interim review of toxicity by sex after 40 evaluable patients/arm; enrolment in female stratum paused if safety threshold crossed.
Feasibility sub-study	First 20 patients: pharmacy preparation lead time; actual vs target infusion start; nursing acceptability; patient preference for scheduling.

are particularly compatible with chronotherapy because they provide 6–7 opportunities for chronotype-adjusted dosing versus only 3 for the three-weekly regimen.

790 *What chronotherapy uniquely offers* is a mechanism that operates *without dose reduction, without additional pharmacological agents, and without modification to the radiotherapy workflow*: the dose, formulation, and schedule are unchanged; only the clock-time of delivery is adjusted. If a 20–25% reduction in Grade ≥ 3 combined toxicity can be achieved through scheduling alone, the clinical argument for its routine adoption — particularly in
795 combination with hyperhydration and radiotherapy optimisation — would be compelling.

This conceptual framework is intentionally modest: it seeks to establish whether chronotype-adjusted delivery reduces toxicity in a measurable way before any claim about efficacy. If a 20–25% reduction in Grade ≥ 3 combined toxicity can be demonstrated while preserving dose intensity, the case for a phase III survival trial — incorporating the
800 sex-stratification lessons of Giacchetti [61] and the circadian monitoring infrastructure of Innominato et al. [65] — becomes credible and clinically compelling.

8. Conclusion

Cisplatin remains the most important chemotherapeutic agent in the concurrent management of locally advanced OSCC and HNSCC, but its organ-specific toxicities — nephro-
805 toxicity, ototoxicity, oral mucositis, nausea and vomiting, and peripheral neuropathy — constitute the principal barrier to treatment completion and full-dose delivery. These toxicities are not interchangeable: each reflects a distinct biology of transporter-mediated platinum uptake, localised adduct formation, and organ-specific apoptotic response. Dose reduction and schedule modification address the aggregate toxicity signal but ignore the
810 mechanistic specificity that makes each organ differently vulnerable.

The circadian molecular clock is documented to regulate, in a tissue-specific manner, at least four of the processes most critical to cisplatin toxicity: OCT2-dependent renal tubular cisplatin influx, XPA-dependent nucleotide excision repair of platinum–DNA adducts, glutathione redox buffering, and mucosal epithelial proliferative rate. The mechanistic
815 case for timing-based toxicity modulation is therefore supported by a body of quantitative molecular data, not by empirical dose-scheduling intuition alone — while remaining *mechanistically plausible* rather than proven in the specific context of OSCC/HNSCC

patient tissues. The BMAL1/ARNTL data reveal a two-tissue asymmetry with tumour-side concordance: both OSCC and NPC studies find that BMAL1 promotes cisplatin sensitivity, while the renal pro-injury role (Zha 2020) awaits independent replication. If confirmed, this host–tumour clock contrast would itself be a targetable feature of chronotherapy — a window in which tumour cisplatin sensitivity is maximised while host uptake and injury are minimised.

The existing clinical evidence for cisplatin chronotherapy in head-and-neck cancer is preliminary: a small OSCC pilot study and a single NPC phase II trial support toxicity-reduction endpoints, but neither is powered for survival, and neither incorporates individual chronotype characterisation. The sex×timing interaction observed in colorectal cancer chronotherapy trials is a mandatory design constraint for future OSCC/HNSCC work. Chronotherapy does not replace established toxicity-reduction strategies — hyperhydration, schedule modification, and radiotherapy optimisation remain the clinical standard — but it occupies a distinct mechanistic niche: it targets cisplatin uptake and repair at the cellular level without dose reduction, pharmacological additives, or changes to the radiotherapy workflow. It is therefore complementary to, not competitive with, existing approaches.

A phase II chronotype-aware trial — using Grade ≥ 3 mucositis or nephrotoxicity within six weeks of first cisplatin as the primary endpoint, with pre-specified sex stratification, prospective CFI monitoring, and biomarker integration — is both scientifically warranted and logistically achievable within standard chemoradiotherapy delivery infrastructure. Whether timing can reduce the toxicity burden enough to improve dose completion rates and, ultimately, survival outcomes in this disease is the question that such a trial is positioned to answer.

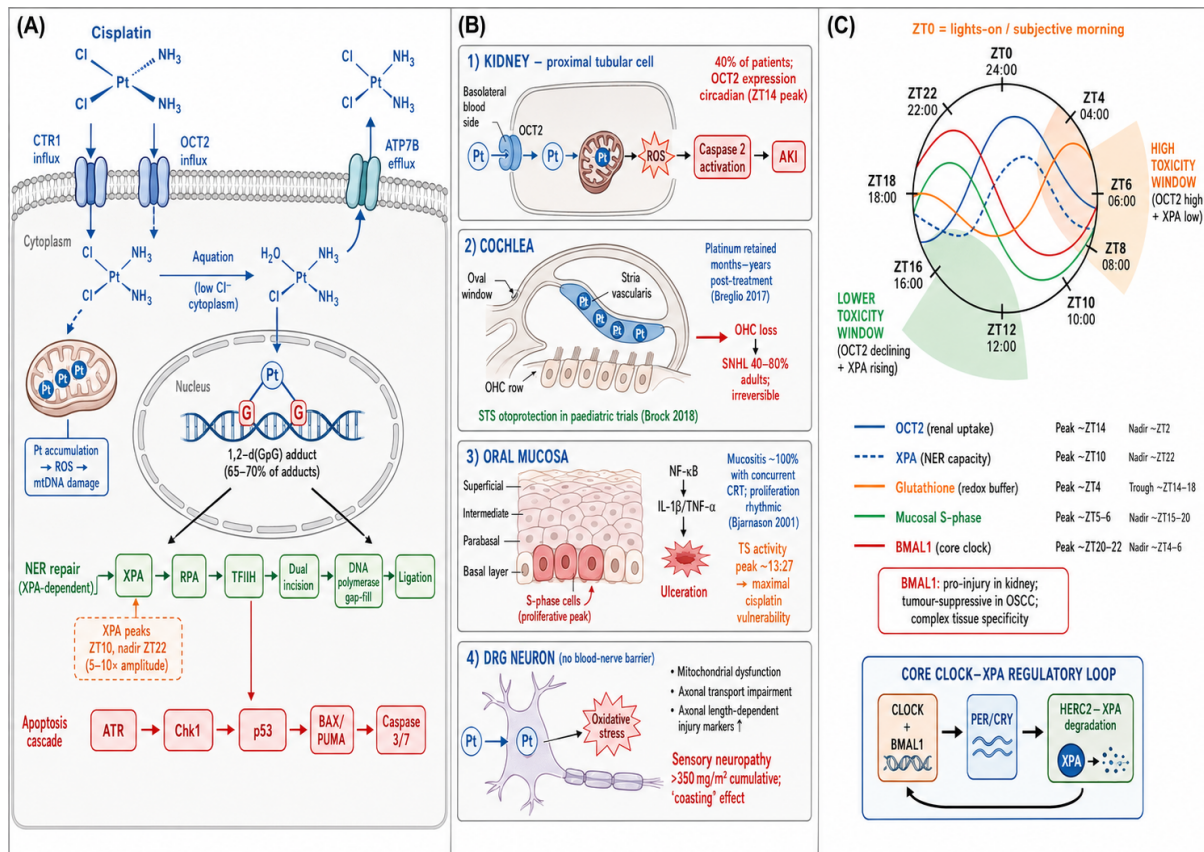


Figure 1: Cisplatin mechanism of action, organ-specific toxicity pathways, and circadian timing gates.

Figure 1 legend. (A) Cellular mechanism of cisplatin cytotoxicity. Cisplatin enters cells via CTR1 (influx) and OCT2 (tissue-specific influx in kidney and cochlea); intracellular export is mediated by ATP7B. Aquation in the low-chloride cytoplasm generates reactive platinum species that form 1,2-d(GpG) intrastrand crosslinks (65–70% of adducts) on nuclear DNA. The NER pathway (XPA scaffold–RPA–TFIIH – dual incision–gap–fill ligation) resolves adducts when repair capacity is sufficient; when overwhelmed, ATR–Chk1–p53–BAX/PUMA–caspase 3/7 apoptosis is activated. XPA protein abundance oscillates with circadian periodicity (peak ZT10, nadir ZT22; 5–10× amplitude), annotated by the dashed clock box. (B) Organ-specific toxicity compartments. *Kidney:* basolateral OCT2 concentrates cisplatin in proximal tubular S3 cells; ROS generation and caspase 2 activation drive AKI (~40% of patients; OCT2 expression is circadian, peaking ZT14). *Cochlea:* platinum is retained in stria vascularis and outer hair cells for months to years post-treatment [33]; irreversible SNHL affects 40–80% of adults; STS otoprotection shown in paediatric trials [35]. *Oral mucosa:* radiation and cisplatin synergise to activate NF-κB–IL-1β/TNF-α, driving ulceration in nearly all CRT patients; thymidylate synthase (TS) activity peaks at ~13:27, identifying the mid-afternoon proliferative window [49]. *DRG neuron:* platinum accumulates without blood–nerve barrier protection; sensory neuropathy emerges above ~350 mg m^{−2} cumulative with a characteristic “coasting” effect [40]. (C) Circadian clock gates superimposed on a 24-hour dial (ZT0 = lights-on/subjective morning). Five oscillating traces are shown: OCT2 renal uptake (solid blue; peak ZT14), XPA NER capacity (dashed blue; peak ZT10), glutathione redox buffer (green; peak ZT4), Mucosal S-phase (orange; peak ZT10–18), and BMAL1 (core clock; peak ~ZT20–22). A shaded region indicates the **HIGH TOXICITY WINDOW** (OCT2 high + XPA low) from ZT6 to ZT10. A **LOWER TOXICITY WINDOW** (OCT2 declining + XPA rising) is shown from ZT12 to ZT16. A box notes: **BMAL1:** pro-injury in kidney; tumour-suppressive in OSCC; complex tissue specificity.

CORE CLOCK–XPA REGULATORY LOOP: CLOCK + BMAL1 → PER/CRY → HERC2–XPA degradation → XPA.

860 trough resting phase), mucosal S-phase proliferation (orange; peak ZT5–6), and BMAL1 mRNA (red;
peak early subjective night). Amber shading marks the *high-toxicity window* (ZT6–8: rising OCT2, low
XPA); teal shading marks the *lower-toxicity window* (ZT14–20: declining OCT2, rising XPA). Inset:
CLOCK/BMAL1–PER/CRY–HERC2-XPA degradation feedback loop. BMAL1 annotation reflects
tissue-specific duality (Section 5). Abbreviations: AKI, acute kidney injury; CTR1, copper transporter 1;
865 DRG, dorsal root ganglion; NER, nucleotide excision repair; OHC, outer hair cell; OCT2, organic cation
transporter 2; ROS, reactive oxygen species; SNHL, sensorineural hearing loss; STS, sodium thiosulfate;
TS, thymidylate synthase; XPA, xeroderma pigmentosum complementation group A; ZT, Zeitgeber time.

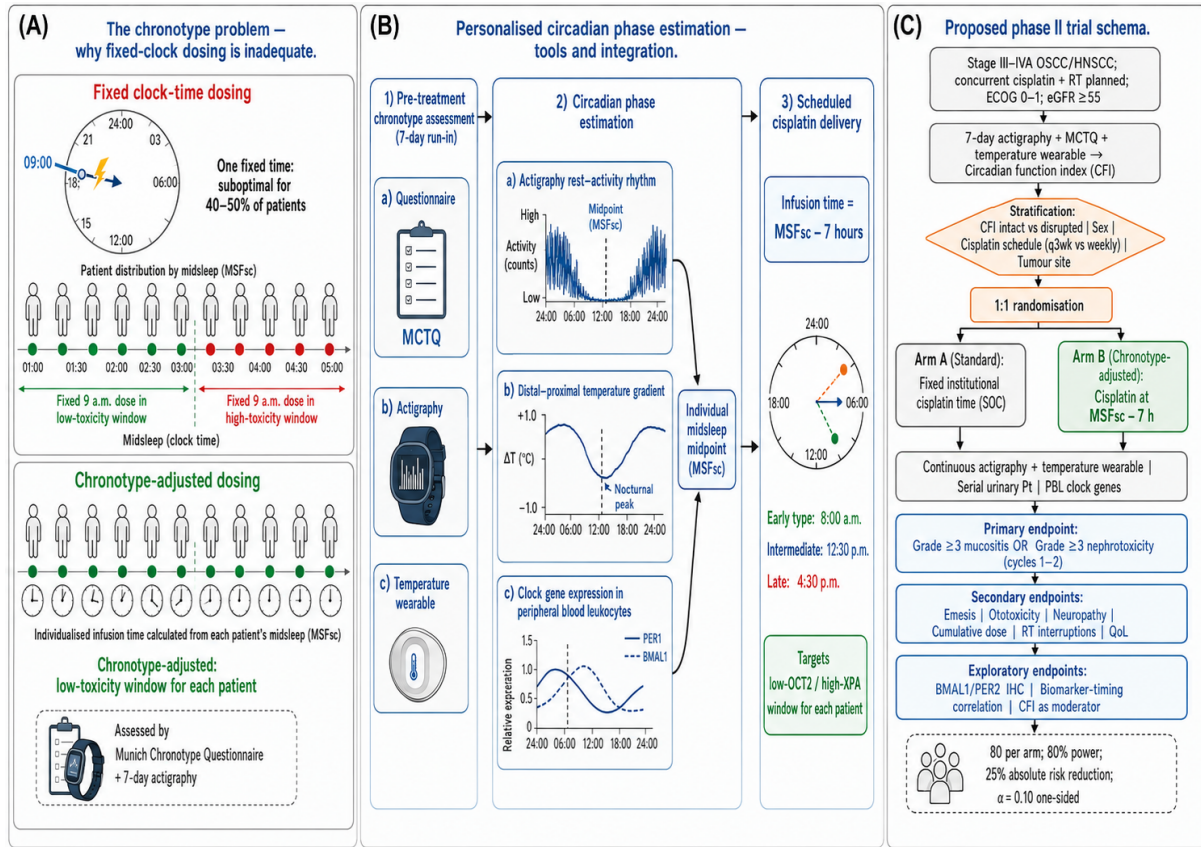


Figure 2: Proposed framework for chronotype-aware cisplatin scheduling in OSCC/HNSCC: rationale, phase estimation, and trial design.

Figure 2 legend. (A) The chronotype problem. *Upper left:* Under fixed institutional clock-time dosing (e.g., 09:00), one single infusion time is applied to all patients regardless of their individual circadian phase. Patient icons are distributed by midsleep on free days (MSFsc; x-axis): early chronotypes (green; midsleep 01:00–02:30) receive cisplatin in their low-toxicity window, whereas late chronotypes (red; midsleep 03:30–05:00) receive it during their high-toxicity window — suboptimal for 40–50% of patients. *Lower left:* Chronotype-adjusted dosing individualises the infusion time so that every patient receives cisplatin in their personal low-toxicity window, assessed by the Munich Chronotype Questionnaire (MCTQ) and 7-day actigraphy. (B) Personalised circadian phase estimation. Three measurement streams converge to estimate individual midsleep (MSFsc): (a) self-reported MCTQ; (b) wrist actigraphy rest-activity rhythm (midpoint of inactivity marked); and (c) continuous distal-proximal skin temperature gradient (nocturnal peak used as phase anchor), plus timed peripheral blood leukocyte (PBL) clock gene expression (PER1/BMAL1 at three time-points). The estimated MSFsc determines the infusion time via the formula $\text{Infusion time} = \text{MSFsc} - 7h$, which targets the low-OCT2/high-XPA window for each patient [47, 44]. Worked examples: early chronotype 08:00; intermediate chronotype 12:30; late chronotype 16:30–17:00. (C) Proposed phase II randomised trial schema (Section 7). Eligible OSCC/HNSCC patients (stage III–IVA; concurrent cisplatin + RT; ECOG 0–1; CrCl ≥ 55; baseline audiogram) enter a 7-day actigraphy + MCTQ + temperature wearable run-in to estimate circadian function index (CFI). Patients are stratified by CFI (intact/disrupted), sex, cisplatin schedule (q3wk/weekly), and tumour site, then randomised 1:1

to Arm A (standard fixed institutional time) or Arm B (chronotype-adjusted: MSFsc – 7 h). Continuous actigraphy, temperature wearable, serial urinary platinum excretion, and PBL clock genes are collected throughout treatment. Primary endpoint: Grade ≥ 3 mucositis or Grade ≥ 3 nephrotoxicity within 6 weeks
890 of first cisplatin (CTCAE v5.0). Secondary endpoints: emesis, ototoxicity, neuropathy, cumulative cisplatin dose, RT interruptions, QoL (EORTC QLQ-C30/ H&N35). Exploratory: BMAL1/PER2 IHC, biomarker-timing correlation, CFI as moderator. Sample size: 80/arm (80% power, primary design target ARR 20%, $\alpha = 0.10$ one-sided). Abbreviations: CFI, circadian function index; CrCl, creatinine clearance; CTCAE, Common Terminology Criteria for Adverse Events; MCTQ, Munich Chronotype Questionnaire;
895 MSFsc, midsleep on free days (sleep-corrected); OCT2, organic cation transporter 2; PBL, peripheral blood leukocytes; QoL, quality of life; RT, radiotherapy; XPA, xeroderma pigmentosum complementation group A.

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